Oxygen and Hydrogen Isotopes: An Analysis of Data from Turkey Lakes and Burlington

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Abstract

The study of oxygen and hydrogen stable isotopes provides important insights into the hydrogeological and climatic controls that influence precipitation. This study examines the isotopic composition of precipitation collected monthly from 1995–2006 at two locations in Ontario: Turkey Lakes and Burlington. Local Meteoric Water Lines (LMWLs) for each site were derived and compared with the Global Meteoric Water Line (GMWL). Turkey Lakes exhibits a LMWL of $\delta D = 7.1849\delta^{18}O + 2.9379$, while Burlington's is $\delta D = 6.3001\delta^{18}O - 6.4412$. The smaller slope and negative intercept in Burlington indicate stronger kinetic effects associated with the condensation processes. Deuterium excess ($d = \delta D - 8\delta^{18}O$) was used to interpret influences such as humidity and evaporation cycles. Turkey Lakes shows stronger relationships between d-excess, relative humidity, and precipitation amount, while Burlington demonstrates a weaker, more variable relationship. Together, these differences highlight how kinetic fractionation shapes isotopic patterns in Southern Ontario precipitation. Understanding these controls is a very important step in identifying large scale hydrogeological processes within the Great Lakes systems.

Introduction

This project will analyze isotopic data collected by Dr. Fred Longstaffe and his team. Data collection occurred monthly from April 1995 to December 2006 by collecting precipitation in a method that ensured no evaporation after collection took place (Longstaffe, 2025). Data collected and provided for the purpose of this report include $\delta^{18}O$, δD, deuterium excess, relative humidity, temperature, and monthly precipitation. Both $\delta^{18}O$ and δD values are standardized to VSMOW. The isotope data analyzed for this project will be from two locations, Turkey Lakes and Burlington (Figure 1). The below equations will be used in the calculations included in this report.

$$(1)\ \delta D = 8 * \delta^{18}O + 10$$

$$(2)\ d = \delta D - 8 * \delta^{18}O$$

Equation one is the Global Meteoric Water Line equation. The equation was published in 1961 and describes the global linear relationship between δD and $\delta^{18}O$ in natural meteoric waters (Craig, 1961). Local deviations from this line occur when kinetic processes influence the isotopic composition of precipitation.

Equation two is used to calculate the deuterium excess. Deuterium excess is a second order isotope parameter defined by the relationship between δD and $\delta^{18}O$ (Merlivat and Jouzel, 1979). It quantifies deviations from the global meteoric water line and is mainly sensitive to humidity. This is because deuterium is less sensitive to kinetic fractionation. Therefore, kinetic fractionation does not deplete the deuterium, and it exists as excess amounts in the vapour. High deuterium excess values typically indicate evaporation under low relative humidity, whereas

lower or negative values suggest enhanced kinetic effects during secondary evaporation (Merlivat and Jouzel, 1979).

Analysis and Interpretation

Turkey Lakes

Meteoric Water Line

The local meteoric water line calculated from all of isotopic data from 1995 to 2006 at Turkey Lakes is $\delta D = (7.1849)\delta^{18}O + 2.9379$ (Figure 2). The local meteoric line has a slightly different slope and a noticeably different intercept than the GMWL. LMWL slopes vary from GMWL slopes due to nonequilibrium processes. These processes can be understood by calculating and analyzing the Deuterium excess (Putman et al., 2019).

Deuterium Excess

Figure 3 shows how the deuterium excess value evolved over the eleven-year observation period. There is a pattern of deuterium excess values rising as they approach colder months (most seen in October and November) and then falling around March and April. However, these trends have a lot of variation and are not consistent. To better understand the averages, the variations, and the anomalies we need to look to the environmental conditions.

Effects

Temperature, relative humidity, and precipitation amount are environmental factors that control equilibrium processes in isotope fractionation. We can use the recorded values for each of these parameters at our study site to make inferences about what causes the observed isotopic fractionation patterns. Figure 4A confirms that the temperature changes associated with seasonal fluctuations do not tell the whole story when it comes to the variations at Turkey Lakes. The trends for relative humidity and deuterium excess follow very similar patterns (Figure 4B). The Deuterium excess anomalies can mostly be explained by anomalies in precipitation amount (Figure 4C). For example, the high deuterium values in the spring of 2001 and the Winter of 2004 loosely correspond with higher precipitation amounts in the months before. Similarly, distinctly low deuterium excess value in October 2003 corresponds with a drop in precipitation. This relationship could possibly be attributed to the amount effect. The amount effect describes the fact that the amount of precipitation has a strong control of isotopes and that large rainfall events can deliver specific isotopic signatures (Vystavna et al., 2021). However, it is important to note that the isotopic signatures produced by the amount effect are not uniform around our world. Interactions with moisture source, recycled vapour, and atmospheric processes can change the strength of this relationship over time (Vystavna et al., 2021). This is important for Ontario, as the site is not a tropical monsoon location. The amount-effect signals may appear in some years or seasons and be muted in others, depending on which moisture source dominates at the time (Vystavna et al., 2021). Therefore, while the deuterium excess precipitation link at Turkey

Lakes supports an amount effect style explanation, we also need to be cautious and consider that the strength of this effect may shift depending on moisture and atmosphere circulation (Vystavna et al., 2021).

Burlington

There are gaps in our data for Burlington due to issues with collection. All calculations and interpretations will be done given the provided data but note that when coming to our conclusions, we may be missing information that could help us determine the processes responsible for the results we see in Burlington.

Meteoric Water Line

The local meteoric water line calculated from all of isotopic data from 1995 to 2006 at Burlington is $\delta D = (6.3001)\delta^{18}O - 6.4412$ (Figure 5). This slope shows a slope that is sligtly different and an intercept that is drastically different that the GMWL.

Deuterium Excess

To account for the difference in GMWL and LMWL, we again can look at the Deuterium excess. The Deuterium excess values for Burlington display a faint pattern of rising around August and dropping around April (Figure 6). Just like for Tukey Lakes, we can look to the environmental conditions to understand anomalies and variations in the deuterium excess data.

Effects

The temperature (Figure 7A) follows an inversely proportional pattern. The deuterium excess rises with temperatures fall and vise versa. The relative humidity (Figure 7B) does not appear to have much correlation with the deuterium excess. A few similarities in the trends for the precipitation amount (Figure 7C) can be observed. Overall, there is no smoking gun for what is causing anomalies in the deuterium excess such as the negative values seen in April 1997, April 1999 and June 2004.

Comparing the Sites

When it comes to comparing the isotopic data from Burlington and Turkey Lakes, the first thing to note is that the Burlington LMWL deviates more from the GMWL than the Turkey Lakes LMWL (Figure 2, Figure 5). Burlington has a much lower slope (6.3001 vs 7.1849 at Turkey Lakes) and also has a strongly negative intercept. The LMWL at Turkey lakes being much closer to the GMWL suggests that the precipitation has undergone more kinetic effects. Kinetic effects occur when clouds experience secondary evaporation (Jia et al., 2022). This leads to the vapour being enriched in heavy isotopes and the deuterium excess being reduced, which can be observed in our results as Turkey Lakes has a significantly lower deuterium excess than Burlington (Figure 8).

The deuterium excess patterns also differ significantly between the two sites (Figure 8). Turkey Lakes shows a stronger connection to relative humidity (figure 4B). Turkey Lakes has a higher relative humidity than Burlington throughout this study (Figure 9B). Kinetic effects are higher in areas with lower relative humidities (Merlivat and Jouzel, 1979). This means there will be higher Deuterium excess values. The higher humidity in Turkey Lakes supports that the lower deuterium excess values are strongly connected to the kinetic effects. In Burlington, humidity has no obvious correlation with deuterium excess. This means that the humidity is not controlling the isotopic fractionation in the precipitation arriving at Burlington.

Discussion

The isotopic variability observed between Turkey Lakes and Burlington fits broader patterns of mid-latitude precipitation in North America. Latitude acts as a control on $\delta^{18}O$ and δD patterns (Putman et al., 2019). A major difference between the two sites is the Y intercept of the LMWL (Turkey Lakes at 2.9 and Burlington at -6). Typical intercept values for continental precipitations at latitudes of around 40-50° exist within a range of around -7 to 15 (Figure 10). Although we do not see the pattern of decreasing intercepts with increasing latitudes (Burlington is at a lower latitude, but exhibits a lower intercept) both intercept values fit what we would expect for continental precipitation at the specified latitude.

The position within the continent also contributes to these isotopic differences. As moist air masses travel inland, Rayleigh distillation progressively depletes heavy isotopes, producing more negative δ values and higher d-excess (Galewsky et al., 2016). Turkey Lakes receives precipitation that has already undergone substantial Rayleigh distillation during its continental trajectory. The Great Lakes act as vast evaporative surfaces that add kinetic fractionation signatures to the precipitated water and therefore influence fractionation processes (Corcoran et al., 2019).

These results demonstrate how latitude, continental position, and kinetic fractionation combine to determine isotopic behavior in mid-latitude continental settings. The Great Lakes amplify these effects by supplying isotopically enriched vapor during evaporation events.

Conclusion

This study shows how isotopic variation in precipitation reflects local environmental conditions in Southern Ontario. The LMWL at Turkey Lakes ($\delta D = 7.1849\delta^{18}O + 2.9379$) is closer to the GMWL, while Burlington's ($\delta D = 6.3001\delta^{18}O – 6.4412$) shows stronger kinetic fractionation. Turkey Lakes exhibits clearer relationships between deuterium excess, humidity, and precipitation amount, suggesting a stronger influence of equilibrium fractionation. Burlington's weaker correlations and lower slope indicate greater secondary evaporation and variability in atmospheric conditions.

These differences show how factors like humidity, moisture source, and position in the continent shape isotopic signatures. Turkey Lakes and Burlington illustrate that even within one climatic region, precipitation isotopes are highly sensitive to local environmental processes and kinetic effects, providing valuable insight into how regional climatic and hydrogeological conditions influence isotope composition in the Great Lakes.

Figures



Figure 1: Map of the Great Lakes Showing Data Collection Locations

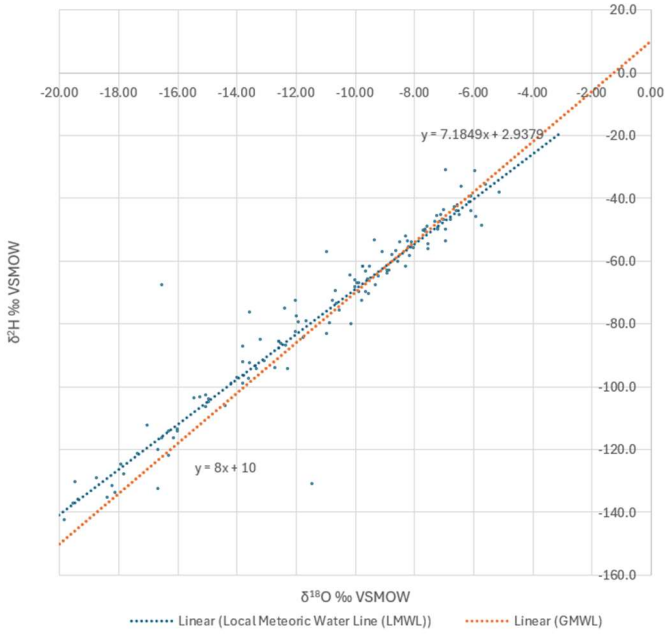


Figure 2: GMWL (Orange) and LMWL (Blue) for Turkey Lakes

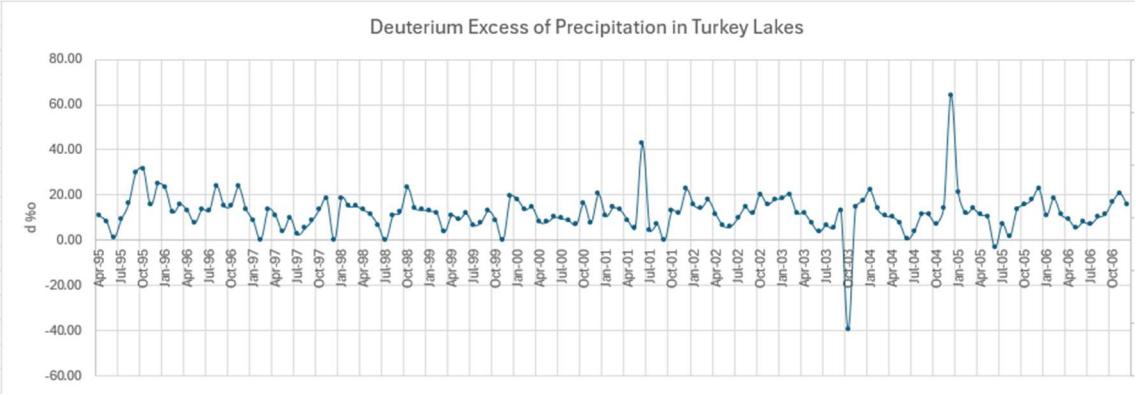


Figure 3: Monthly d excess for Turkey Lakes from 1995-2006

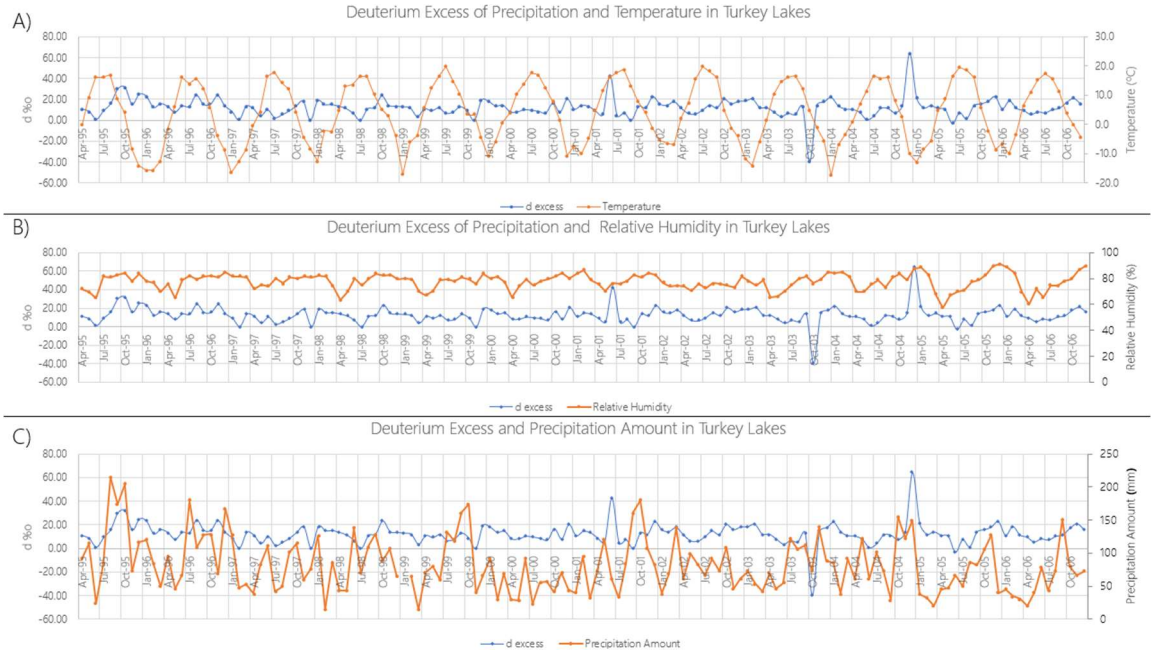


Figure 4: Deuterium Excess and (A) Temperature (B) Relative Precipitation (C) Precipitation Amount in Turkey Lakes from 1995-2006

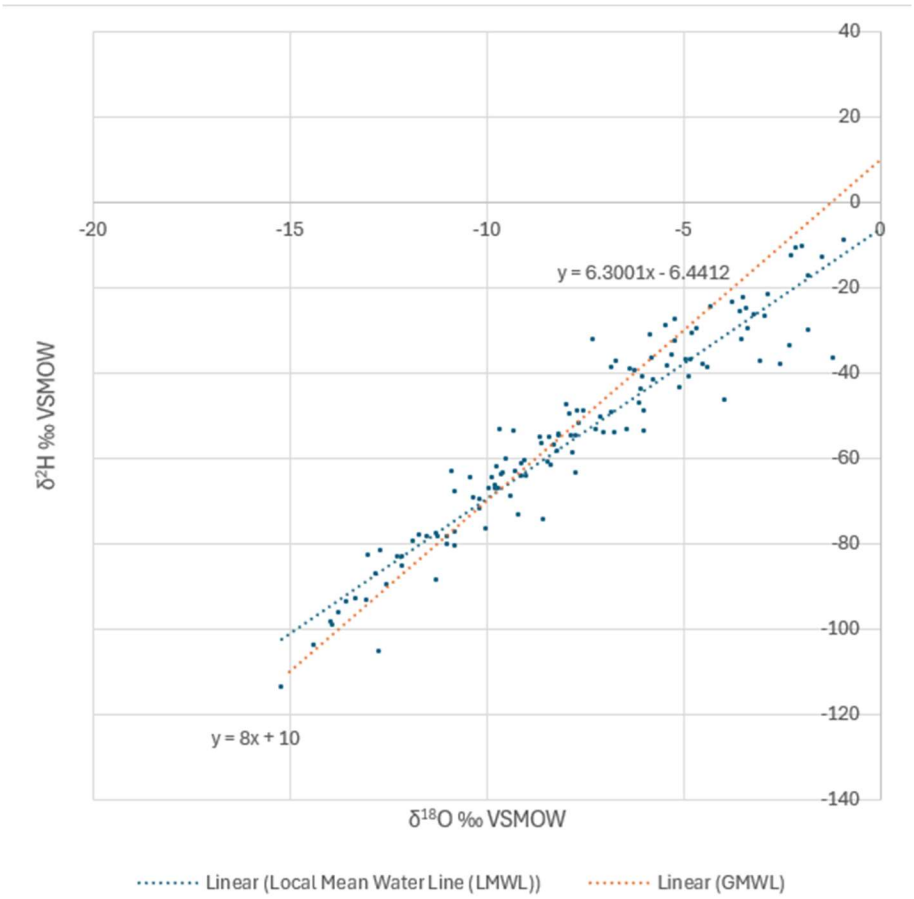


Figure 5: LMWL (Blue) and GMWL (Orange) of Burlington

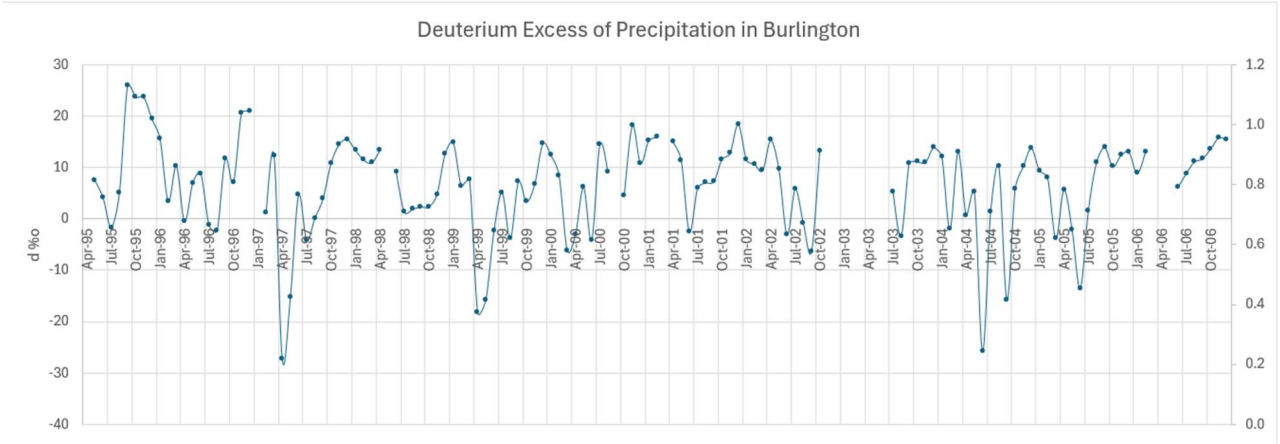


Figure 6: Deuterium Excess for Burlington from 1995-2006

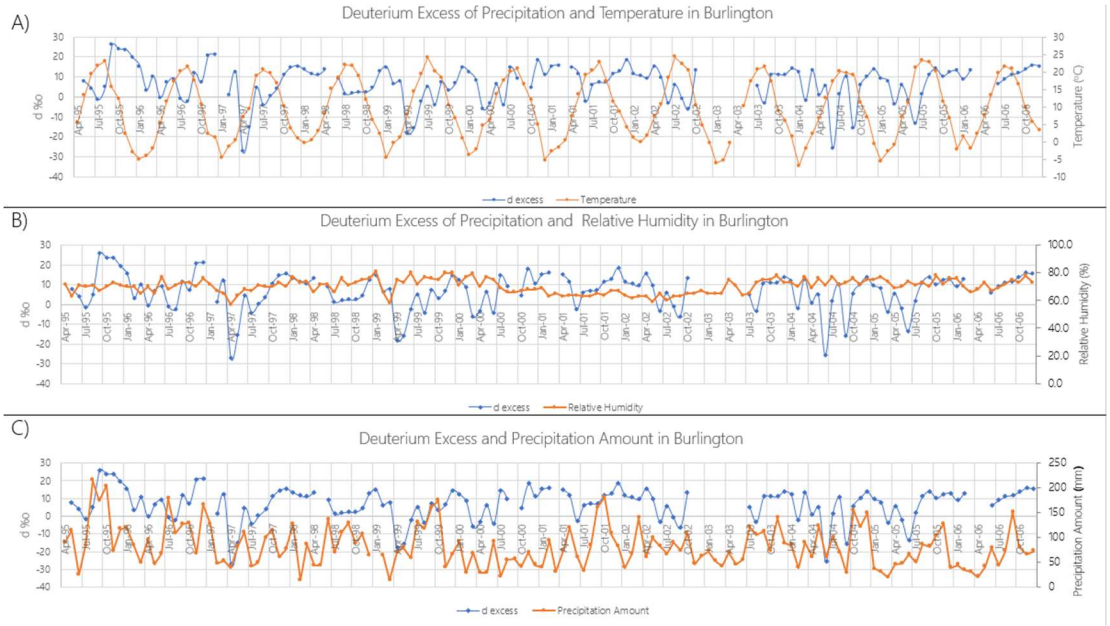


Figure 7: Deuterium Excess and (A) Temperature (B) Relative Humidity (C) Precipitation Amount in Burlington from 1995-2006

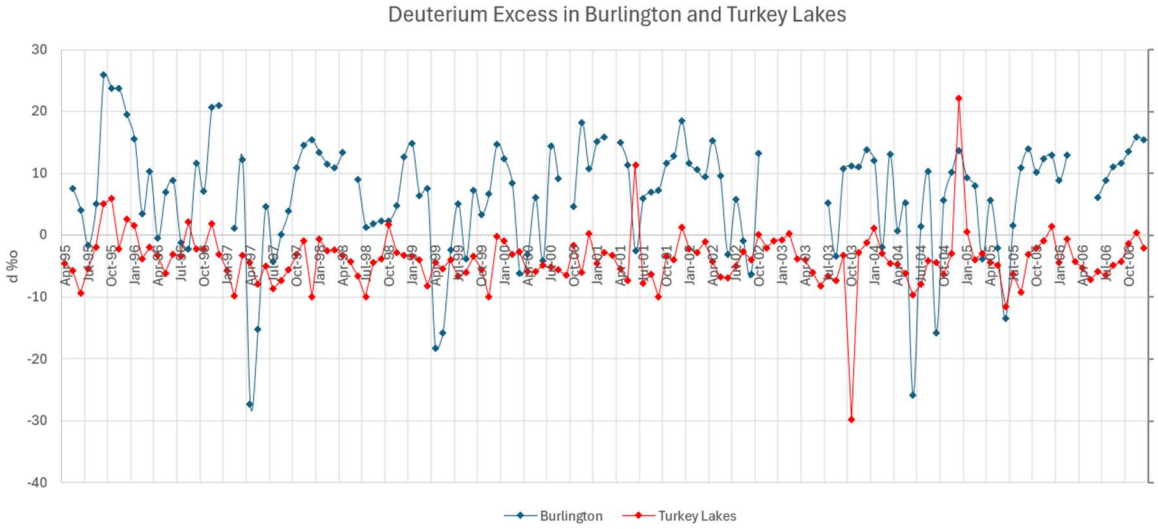


Figure 8: Deuterium Excess in Burlington and Turkey Lakes

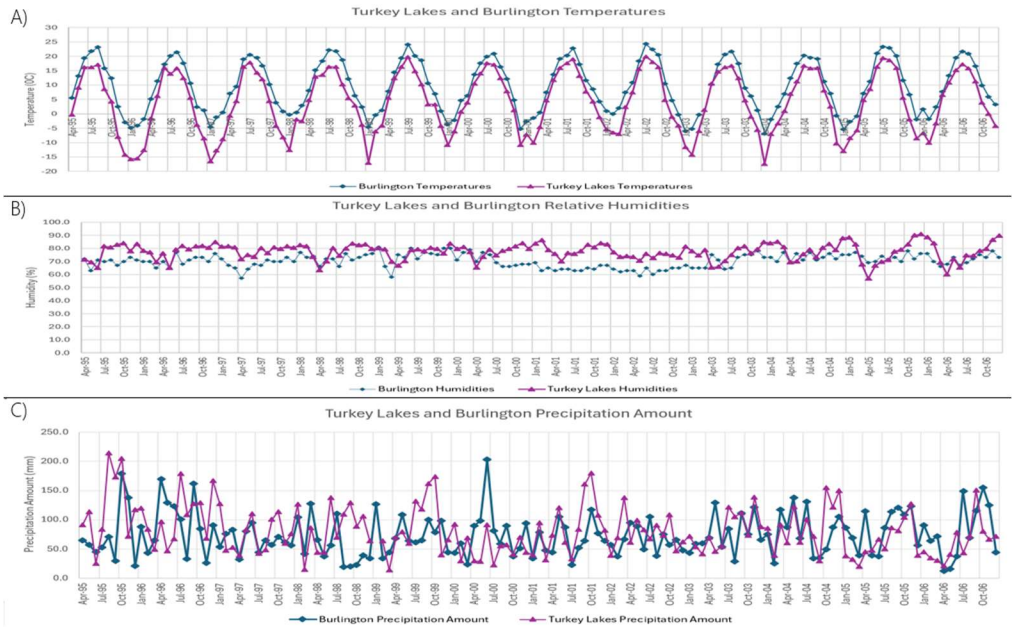


Figure 9: Turkey lakes and Burlington (A) Temperatures (B) Relative Humidities and (C) Precipitation Amount

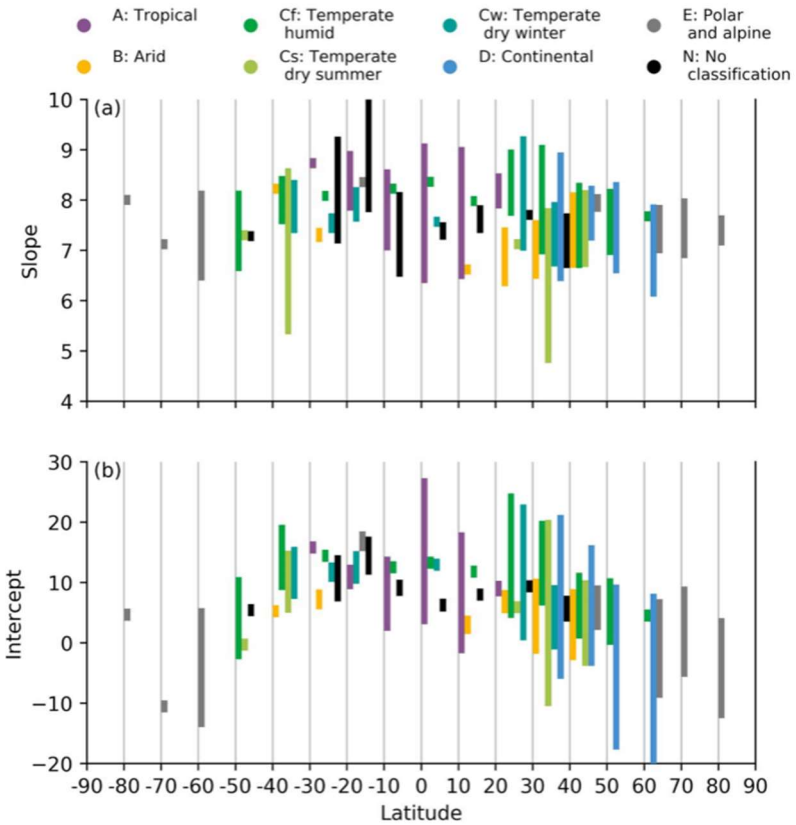


Figure 10. Distribution of (a) slopes and (b) intercepts of local meteoric water lines for different latitudes and climatic conditions. Retrieved from (Putman et al., 2019)

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